

Somaiya Vidyavihar University
K J Somaiya School of Engineering
Department of Electronics and Telecommunication Engineering



SOMAIYA
VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya Vidyavihar University
K J Somaiya School of Engineering



Department of Electronics and Telecommunication Engineering

Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 04
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

[Signature]
10/9/26

Experiment No: 4

Title: : DML – select, insert, update and delete

1. Group by, having clause, aggregate functions, Set Operations
2. Nested queries : AND, OR, NOT, IN, NOT IN, Exists, Not Exists, Between, Like, Alias, ANY, ALL, DISTINCT
3. Update
4. Delete

Objective: Define/modify database definitions with proper constraints

OUTCOME:

- CO3. Understand the internal storage structures in a physical DB design

Books/ Journals/ Websites referred:

1. G. K. Gupta : "Database Management Systems", McGraw – Hill
2. Korth, Silberchatz, Sudarshan : "Database Systems Concept", 6th Edition , McGraw Hill
3. Elmasri and Navathe, "Fundamentals of Database Systems", 5th Edition, PEARSON Education.

Resources used: Postgres

Theory:

Select: The SQL **SELECT** statement is used to fetch the data from a database table which returns this data in the form of a result table. These result tables are called result-sets.

Syntax

The basic syntax of the **SELECT** statement is as follows –

SELECT column1, column2, columnN **FROM** table_name;

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The basic syntax of the **SELECT** statement is as follows –

SELECT column1, column2, columnN **FROM** table_name;

Here, column1, column2... are the fields of a table whose values you want to fetch. If you want to fetch all the fields available in the field, then you can use the following syntax.

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SELECT * FROM table_name;

The following code is an example, which would fetch the ID, Name and Salary fields of the customers available in CUSTOMERS table.

SQL> SELECT ID, NAME, SALARY FROM CUSTOMERS;

Insert: The SQL **INSERT INTO** Statement is used to add new rows of data to a table in the database.

i. Syntax

There are two basic syntaxes of the INSERT INTO statement which are shown below.

INSERT INTO TABLE_NAME (column1, column2, column3,...columnN)

VALUES (value1, value2, value3,...valueN);

ii. Example

The following statements would create record in the CUSTOMERS table.

INSERT INTO CUSTOMERS (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (1, 'Ramesh', 32, 'Ahmedabad', 2000.00);

Update: The SQL **UPDATE** Query is used to modify the existing records in a table. You can use the WHERE clause with the UPDATE query to update the selected rows, otherwise all the rows would be affected.

Syntax:

The basic syntax of the UPDATE query with a WHERE clause is as follows –

UPDATE table_name

SET column1 = value1, column2 = value2....., columnN = valueN

WHERE [condition];

You can combine N number of conditions using the AND or the OR operators.

The following query will update the ADDRESS for a customer whose ID number is 6 in the table.

SQL> UPDATE CUSTOMERS
SET ADDRESS = 'Pune'
WHERE ID = 6;

Delete: The SQL **DELETE** Query is used to delete the existing records from a table.

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You can use the WHERE clause with a DELETE query to delete the selected rows, otherwise all the records would be deleted.

Syntax

The basic syntax of the DELETE query with the WHERE clause is as follows –

DELETE FROM table_name

WHERE [condition];

The following code has a query, which will DELETE a customer, whose ID is 6.

**SQL> DELETE FROM CUSTOMERS
WHERE ID = 6;**

Clauses and Operators

1. **Group by clause:** These are circumstances where we would like to apply the aggregate functions to a single set of tuples but also to a group of sets of tuples we would like to specify this wish in SQL using the group by clause. The attributes or attributes given by the group by clause are used to form groups. Tuples with the same value on all attributes in the group by clause placed in one group.

Example:.

```
Select<attribute_name,avg(<attribute_name>)as  
<new_attribute_name>| From <table_name>  
Group by <attribute_name>
```

**Example: select designation, sum(salary) as total_salary from employee group by
Designation;**

2. **Having clause:** A having clause is like a where clause but only applies only to groups as a whole whereas the where clause applies to the individual rows. A query can contain both where clause and a having clause. In that case

a. The where clause is applied first to the individual rows in the tables or table structures objects in the diagram pane. Only the rows that meet the conditions in the where clause are grouped.

b. The having clause is then applied to the rows in the result set that are produced by grouping. Only the groups that meet the having conditions appear in the query output.

Example:

```
select dept_no from EMPLOYEE group_by dept_no
```

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**having avg (salary) >=all (select avg (salary)
from EMPLOYEE group by dept_no);**

3. Aggregate functions: Aggregate functions such as SUM, AVG, count, count (*), MAX and MIN generate summary values in query result sets. An aggregate functions (with the exception of count (*)) processes all the selected values in a single column to produce a single result value

**Example: select dept_no,count (*)
from EMPLOYEE group by dept_no;**

Example: select max (salary)as maximum from EMPLOYEE;

Example: select sum (salary) as total_salary from EMPLOYEE;

Example: Select min (salary) as minsal from EMPLOYEE;

4. Exists and Not Exists: Subqueries introduced with exists and not queries can be used for two set theory operations: Intersection and Difference. The intersection of two sets contains all elements that belong to both of the original sets. The difference contains elements that belong to only first of the two sets.

Example:

**Select *from DEPARTMENT
where exists(select * from PROJECT
where DEPARTMENT.dept_no = PROJECT.dept_no) ;**

5. IN and Not In: SQL allows testing tuples for membership in a relation. The “in” connective tests for set membership where the set is a collection of values produced by select clause. The “not in” connective tests for the absence of set membership. The in and not in connectives can also be used on enumerated sets.

Example:

- 1. Select fname, mname, lname from employee where designation In (“ceo”, “manager”, “hod”, “assistant”)**
- 2. Select fullname from department where relationship not in(“brother”);**

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6. Between: The BETWEEN operator selects values within a given range. The values can be numbers, text, or dates. The BETWEEN operator is inclusive. Begin and end values are included.

Syntax:

SELECT column_name(s)

FROM table_name

WHERE column_name BETWEEN value1 AND value2;

Example:

SELECT * FROM Products WHERE Price BETWEEN 10 AND 20;

7. LIKE: The LIKE **operator** is used in a WHERE clause to search for a specified pattern in a column.

There are two wildcards used in conjunction with the LIKE operator:

- % - The percent sign represents zero, one, or multiple characters
- _ - The underscore represents a single character

Syntax: **SELECT column1, column2, ...**

FROM table_name

WHERE columnN LIKE pattern

Examples:

1. selects all customers with a CustomerName starting with "a":

SELECT * FROM Customers

WHERE CustomerName LIKE 'a%';

2. selects all customers with a CustomerName that have "r" in the second position:

SELECT * FROM Customers

WHERE CustomerName LIKE '_r%';

8. Alias: The use of table aliases is to rename a table in a specific SQL statement. The renaming is a temporary change and the actual table name does not change in the database. The column aliases are used to rename a table's columns for the purpose of a particular SQL query.

The basic syntax of a **table** alias is as follows.

SELECT column1, column2....

FROM table_name AS alias_name

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WHERE [condition];

The basic syntax of a **column** alias is as follows.

SELECT column_name AS alias_name

FROM table_name

WHERE [condition];

Example:

```
SELECT C.ID, C.NAME, C.AGE, O.AMOUNT  
  
FROM CUSTOMERS AS C, ORDERS AS O  
  
WHERE C.ID = O.CUSTOMER_ID;
```

9. Distinct: The SELECT DISTINCT statement is used to return only distinct (different) values.

Syntax: SELECT DISTINCT *column1, column2, ...*
FROM *table_name*;

Example: SELECT DISTINCT Country FROM Customers;

10. Set Operations: 4 different types of SET operations, along with example:

1. UNION
2. UNION ALL
3. INTERSECT
4. MINUS

UNION Operation

UNION is used to combine the results of two or more SELECT statements. However it will eliminate duplicate rows from its resultset. In case of union, number of columns and datatype must be same in both the tables, on which UNION operation is being applied.

Query: **SELECT * FROM First**

UNION

SELECT * FROM Second;

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UNION ALL

This operation is similar to Union. But it also shows the duplicate rows.

Query: **SELECT * FROM First**

UNION ALL

SELECT * FROM Second;

INTERSECT

Intersect operation is used to combine two SELECT statements, but it only returns the records which are common from both SELECT statements. In case of **Intersect** the number of columns and datatype must be same.

Query: **SELECT * FROM First**

INTERSECT

SELECT * FROM Second;

MINUS

The Minus operation combines results of two SELECT statements and return only those in the final result, which belongs to the first set of the result.

Query: **SELECT * FROM First**

MINUS

SELECT * FROM Second;

11. ANY and ALL: The ANY and ALL operators are used with a WHERE or HAVING clause. The ANY operator returns true if any of the subquery values meet the condition. The ALL operator returns true if all of the subquery values meet the condition.

iii.ANY

SELECT *column_name(s)*
FROM *table_name*
WHERE *column_name operator ANY*
(SELECT *column_name* FROM *table_name* WHERE *condition*);

Example: The following SQL statement returns TRUE and lists the productnames if it finds ANY records in the OrderDetails table that quantity = 10:

SELECT ProductName
FROM Products

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WHERE ProductID

= ANY (SELECT ProductID FROM OrderDetails WHERE Quantity = 10);

iv.ALL

SELECT *column_name(s)*

FROM *table_name*

WHERE *column_name operator ALL*

(SELECT *column_name* FROM *table_name* WHERE *condition*);

Example: The following SQL statement returns TRUE and lists the productnames if ALL the records in the OrderDetails table has quantity = 10:

SELECT ProductName

FROM Products

WHERE ProductID = ALL (SELECT ProductID FROM OrderDetails WHERE Quantity = 10);

Problem Statement: Database Management System for "NovaTech Enterprises"

Organization Background: NovaTech Enterprises is a mid-sized organization dealing with employees, departments, products, customers, and sales orders. The management wants to design and query a relational database to analyze employee performance, department budgets, product sales, and customer behavior. As a database developer, you are required to design the schema and write SQL queries to meet the following business requirements.

Suggested Schema

Department (DeptID, DeptName, Location, Budget) **Employee** (EmpID, EmpName, DeptID, Salary, JoiningDate, ManagerID, Email) **Product** (ProductID, ProductName, Category, Price, Stock) **Customer** (CustomerID, CustomerName, City, Country, Phone) **Orders** (OrderID, CustomerID, ProductID, EmpID, Quantity, OrderDate)

1. GROUP BY, HAVING, Aggregate Functions, Set Operations

- Find the **total salary and average salary** paid to employees in each department (use SUM, AVG, GROUP BY DeptID).
- Find the **count of employees** in each department, and display only those departments having **more than 5 employees** (use COUNT, GROUP BY, HAVING).

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- Find the **total quantity of products sold** per product category, showing only categories where total sales exceed 100 units (SUM, GROUP BY, HAVING).
- Find the **maximum and minimum salary** in the organization (MAX, MIN).
- Retrieve a list of **all cities where either employees reside or customers reside** (using UNION).
- Retrieve **customers who have placed orders in both January and February** (using INTERSECT).

2. Nested Queries: AND, OR, NOT, IN, NOT IN, EXISTS, NOT EXISTS, BETWEEN, LIKE, Alias, ANY, ALL, DISTINCT (Any five)

- Display employees whose **salary is between 30,000 and 60,000** and who joined **after 2020** (BETWEEN, AND).
- Find employees who work in the 'HR' department **OR have a salary greater than 50,000** (OR).
- Find employees who are **NOT part of the 'IT' department** (NOT).
- List employees whose **DeptID is IN** the set of departments located in 'Mumbai' or 'Pune' (IN with subquery).
- List products whose **ProductID is NOT IN** the list of products that have never been ordered (NOT IN).
- Find customers **who have placed at least one order** using a correlated subquery (EXISTS).
- Find customers **who have never placed any order** (NOT EXISTS).
- Retrieve all employees whose **name starts with 'A' and ends with 'n'** (LIKE '%A%n' pattern, e.g., LIKE 'A%').
- Display department name and average salary using a **column alias** (e.g., AVG(Salary) AS AvgSalary).
- Find employees whose **salary is greater than ANY** employee's salary in the 'HR' department (> ANY).
- Find employees whose **salary is greater than ALL** employees' salaries in the 'Sales' department (> ALL).
- Retrieve the **DISTINCT list of cities** from which customers have placed orders (DISTINCT).

3. UPDATE

- **Increase the salary by 10%** for all employees in the 'IT' department.
- **Update the stock quantity** of a product after an order is placed (reduce stock by ordered quantity).
- **Change the department location** for a given DeptID.
- **Update customer contact details** (phone/city) for a specific CustomerID.

4. DELETE

- **Delete an employee record** whose EmpID matches a given value (e.g., an employee who has resigned).
- **Delete all orders** placed by a customer who has been marked inactive.

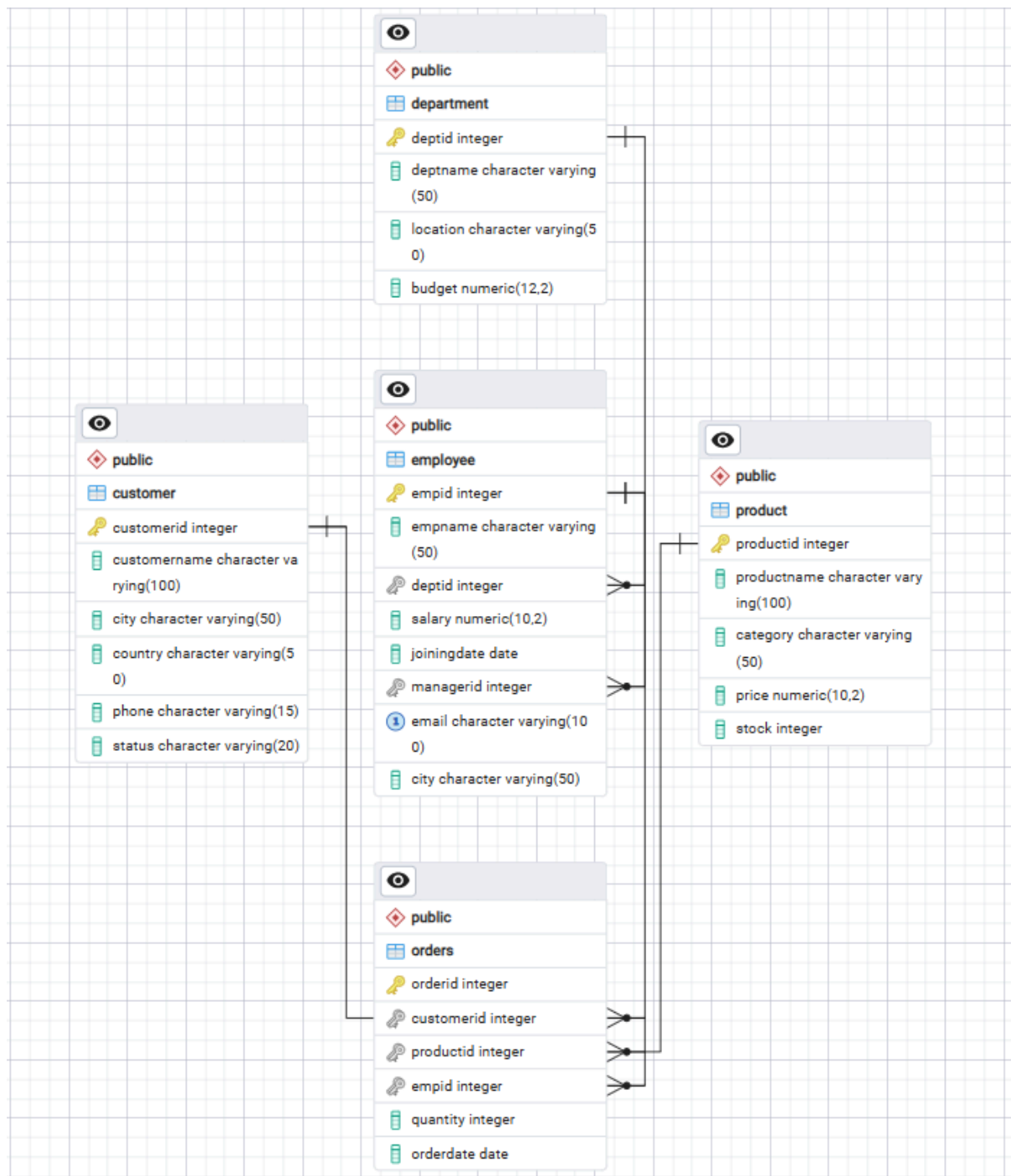
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- **Delete products** from the Product table where stock is **zero and no orders exist** (using a nested `NOT EXISTS` subquery inside `DELETE`).
- **Delete department records** having no employees assigned to them.

Outputs :--

1. ER Diagram and table creation scripts (with primary/foreign keys).

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2. Sample data insertion (at least 10 rows per table).

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VALUES

```
(1, 'HR', 'Mumbai', 900000),
(2, 'IT', 'Pune', 1500000),
(3, 'Sales', 'Mumbai', 1200000),
(4, 'Finance', 'Pune', 1000000),
(5, 'Marketing', 'Mumbai', 800000),
(6, 'Operations', 'Thane', 700000),
(7, 'Admin', 'Navi Mumbai', 600000),
(8, 'R&D', 'Pune', 1300000),
(9, 'Legal', 'Mumbai', 500000),
(10, 'Training', 'Thane', 400000);
```

VALUES

```
(101, 'Arjun', 1, 38000, '2021-06-15', NULL, 'arjun@novatech.com', 'Mumbai'),
(102, 'Aman', 1, 42000, '2022-01-10', 101, 'aman@novatech.com', 'Pune'),
(103, 'Rohan', 1, 36000, '2019-08-20', 101, 'rohan@novatech.com', 'Thane'),
(104, 'Kiran', 1, 47000, '2020-03-12', 101, 'kiran@novatech.com', 'Mumbai'),
(105, 'Ayaan', 1, 52000, '2021-11-05', 101, 'ayaan@novatech.com', 'Navi Mumbai'),
(106, 'Anand', 1, 34000, '2023-02-01', 101, 'anand@novatech.com', 'Pune'),
(107, 'Amit', 2, 65000, '2019-07-11', NULL, 'amit@novatech.com', 'Mumbai'),
(108, 'Priya', 2, 72000, '2021-04-19', 107, 'priya@novatech.com', 'Pune'),
(109, 'Neha', 2, 68000, '2022-09-23', 107, 'neha@novatech.com', 'Thane'),
(110, 'Vikas', 3, 55000, '2018-05-14', NULL, 'vikas@novatech.com', 'Mumbai'),
(111, 'Sneha', 3, 58000, '2020-10-02', 110, 'sneha@novatech.com', 'Pune'),
(112, 'Rahul', 3, 62000, '2021-01-15', 110, 'rahul@novatech.com', 'Mumbai'),
(113, 'Meera', 4, 60000, '2019-12-12', NULL, 'meera@novatech.com', 'Navi Mumbai'),
(114, 'Ishita', 5, 50000, '2022-06-18', NULL, 'ishita@novatech.com', 'Thane'),
(115, 'Kunal', 6, 45000, '2023-01-09', NULL, 'kunal@novatech.com', 'Pune'),
(116, 'Aarav', 7, 70000, '2020-11-21', NULL, 'aarav@novatech.com', 'Mumbai');
```

VALUES

```
(1, 'Laptop', 'Electronics', 65000, 50),
(2, 'Mouse', 'Accessories', 800, 150),
(3, 'Keyboard', 'Accessories', 1500, 120),
(4, 'Monitor', 'Electronics', 15000, 70),
(5, 'Printer', 'Office', 12000, 60),
(6, 'Router', 'Networking', 4500, 80),
(7, 'Switch', 'Networking', 6500, 75),
(8, 'SSD', 'Storage', 5500, 100),
(9, 'Webcam', 'Accessories', 3000, 90),
(10, 'Office Chair', 'Furniture', 8500, 0),
(11, 'Office Desk', 'Furniture', 15000, 0),
(12, 'Projector', 'Office', 35000, 40);
```

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VALUES

```
(1, 'Laptop', 'Electronics', 65000, 50),
(2, 'Mouse', 'Accessories', 800, 150),
(3, 'Keyboard', 'Accessories', 1500, 120),
(4, 'Monitor', 'Electronics', 15000, 70),
(5, 'Printer', 'Office', 12000, 60),
(6, 'Router', 'Networking', 4500, 80),
(7, 'Switch', 'Networking', 6500, 75),
(8, 'SSD', 'Storage', 5500, 100),
(9, 'Webcam', 'Accessories', 3000, 90),
(10, 'Office Chair', 'Furniture', 8500, 0),
(11, 'Office Desk', 'Furniture', 15000, 0),
(12, 'Projector', 'Office', 35000, 40);
```

VALUES

```
(1, 'Raj Enterprises', 'Mumbai', 'India', '9876500001', 'ACTIVE'),
(2, 'Tech Solutions', 'Pune', 'India', '9876500002', 'ACTIVE'),
(3, 'Global Traders', 'Delhi', 'India', '9876500003', 'ACTIVE'),
(4, 'Smart Systems', 'Mumbai', 'India', '9876500004', 'ACTIVE'),
(5, 'Alpha Corp', 'Bangalore', 'India', '9876500005', 'ACTIVE'),
(6, 'Bright Industries', 'Pune', 'India', '9876500006', 'ACTIVE'),
(7, 'Metro Services', 'Chennai', 'India', '9876500007', 'ACTIVE'),
(8, 'Prime Solutions', 'Mumbai', 'India', '9876500008', 'ACTIVE'),
(9, 'Future Tech', 'Hyderabad', 'India', '9876500009', 'ACTIVE'),
(10, 'Old Traders', 'Delhi', 'India', '9876500010', 'INACTIVE');
```

VALUES

```
(1001, 1, 1, 107, 20, '2026-01-05'),
(1002, 1, 2, 110, 40, '2026-01-15'),
(1003, 1, 4, 107, 15, '2026-02-10'),
(1004, 1, 6, 110, 55, '2026-02-18'),

(1005, 2, 3, 111, 35, '2026-01-08'),
(1006, 2, 5, 112, 60, '2026-02-05'),
(1007, 2, 7, 111, 50, '2026-02-20'),

(1008, 3, 8, 107, 80, '2026-03-10'),
(1009, 4, 9, 110, 30, '2026-03-15'),
(1010, 5, 12, 111, 50, '2026-04-01'),
(1011, 6, 2, 112, 35, '2026-04-05'),
(1012, 7, 3, 110, 30, '2026-04-10'),
(1013, 8, 4, 107, 15, '2026-04-15'),

(1014, 10, 1, 107, 5, '2026-05-01'),
(1015, 10, 5, 110, 10, '2026-05-05'),
```

3. SQL queries for all the above points, properly labeled by topic.

GROUP BY + Aggregate Functions

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Query 1 — Total and average salary by department

```
SQL
SELECT

    DeptID,

    SUM(Salary) AS Total_Salary,

    AVG(Salary) AS Average_Salary

FROM Employee

GROUP BY DeptID;
```

Query 2 — Departments having more than 5 employees

```
SQL
SELECT

    DeptID,

    COUNT(*) AS Employee_Count

FROM Employee

GROUP BY DeptID

HAVING COUNT(*) > 5;
```

Query 3 — Total quantity sold per category where sales > 100

```
SQL
SELECT

    p.Category,

    SUM(o.Quantity) AS Total_Quantity

FROM Product p

JOIN Orders o
```

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```
ON p.ProductID = o.ProductID  
  
GROUP BY p.Category  
  
HAVING SUM(o.Quantity) > 100;
```

Query 4 — Maximum salary

```
SQL  
SELECT MAX(Salary) AS Maximum_Salary  
  
FROM Employee;
```

Query 5 — Minimum salary

```
SQL  
SELECT MIN(Salary) AS Minimum_Salary  
  
FROM Employee;
```

UNION — cities of employees or customers

```
SQL  
SELECT City  
  
FROM Employee  
  
  
UNION  
  
SELECT City  
  
FROM Customer;
```

UNION ALL

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SQL

```
SELECT City  
  
FROM Employee
```

```
UNION ALL
```

```
SELECT City  
  
FROM Customer;
```

INTERSECT — customers who ordered in both January and February

SQL

```
SELECT CustomerID  
  
FROM Orders  
  
WHERE OrderDate >= '2026-01-01'  
      AND OrderDate < '2026-02-01'
```

```
INTERSECT
```

```
SELECT CustomerID  
  
FROM Orders  
  
WHERE OrderDate >= '2026-02-01'  
      AND OrderDate < '2026-03-01';
```

EXCEPT — PostgreSQL equivalent of MINUS

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SQL

SELECT CustomerID

FROM Customer

EXCEPT

SELECT CustomerID

FROM Orders;

Q6 — BETWEEN + AND

SQL

SELECT *

FROM Employee

WHERE Salary **BETWEEN** 30000 **AND** 60000

AND JoiningDate > '2020-12-31';

Q7 — OR

SQL

SELECT *

FROM Employee e

JOIN Department d

ON e.DeptID = d.DeptID

WHERE d.DeptName = 'HR'

OR e.Salary > 50000;

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Q8 — NOT

```
SQL
SELECT *
FROM Employee e
JOIN Department d
ON e.DeptID = d.DeptID
WHERE NOT d.DeptName = 'IT';
```

Q9 — IN with subquery

```
SQL
SELECT *
FROM Employee
WHERE DeptID IN
(
    SELECT DeptID
    FROM Department
    WHERE Location IN ('Mumbai', 'Pune')
);
```

Q10 — NOT IN

```
SQL
SELECT *
FROM Product
WHERE ProductID NOT IN
```

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```
(  
    SELECT ProductID  
    FROM Product  
    WHERE ProductID NOT IN  
    (  
        SELECT ProductID  
        FROM Orders  
    )  
);
```

Q20 — Increase IT salary by 10%

```
SQL  
UPDATE Employee  
SET Salary = Salary * 1.10  
WHERE DeptID =  
(  
    SELECT DeptID  
    FROM Department  
    WHERE DeptName = 'IT'  
);
```

```
SQL  
SELECT *
```

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```
FROM Employee  
WHERE DeptID = 2;
```

Q21 — Reduce product stock after order

```
SQL  
UPDATE Product  
SET Stock = Stock - 5  
WHERE ProductID = 1;
```

```
SQL  
SELECT *  
FROM Product  
WHERE ProductID = 1;
```

Q22 — Change department location

```
SQL  
UPDATE Department  
SET Location = 'Mumbai'  
WHERE DeptID = 4;
```

```
SQL  
SELECT *  
FROM Department
```

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```
WHERE DeptID = 4;
```

Q23 — Update customer contact details

```
SQL  
UPDATE Customer  
  
SET Phone = '9999999999',  
  
    City = 'Pune'  
  
WHERE CustomerID = 5;
```

```
SQL  
SELECT *  
  
FROM Customer  
  
WHERE CustomerID = 5;
```

Q24 — Delete employee by EmpID

```
SQL  
DELETE FROM Employee  
  
WHERE EmpID = 116;
```

```
SQL  
SELECT * FROM Employee;
```

Q25 — Delete orders of inactive customer

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SQL

```
DELETE FROM Orders  
  
WHERE CustomerID IN  
  
(  
  
    SELECT CustomerID  
  
    FROM Customer  
  
    WHERE Status = 'INACTIVE'  
  
);
```

SQL

```
SELECT *  
  
FROM Orders  
  
WHERE CustomerID = 10;
```

DELETE using nested NOT EXISTS

SQL

```
DELETE FROM Product p  
  
WHERE p.Stock = 0  
  
AND NOT EXISTS  
  
(  
  
    SELECT 1  
  
    FROM Orders o
```

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```
WHERE o.ProductID = p.ProductID  
);
```

```
SQL  
SELECT *  
FROM Product;
```

Q26 — Delete Departments having no employees

```
SQL  
DELETE FROM Department d  
WHERE NOT EXISTS  
(  
    SELECT 1  
    FROM Employee e  
    WHERE e.DeptID = d.DeptID  
);
```

```
SQL  
SELECT *  
FROM Department;
```

4. Output screenshots/results for each query.

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Q1

	deptid integer	total_salary numeric	average_salary numeric
1	3	175000.00	58333.333333333333
2	5	50000.00	50000.000000000000
3	4	60000.00	60000.000000000000
4	6	45000.00	45000.000000000000
5	2	225500.00	75166.666666666667
6	1	249000.00	41500.000000000000

Q2

	deptid integer	employee_count bigint
1	1	6

Q3

	category character varying (50)	total_quantity bigint
1	Storage	120
2	Accessories	215
3	Networking	210
4	Office	170

Q4

	maximum_salary numeric
1	79200.00

Q5

	minimum_salary numeric
1	34000.00

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	city character varying (50) 
1	Pune
2	Hyderabad
3	Mumbai
4	Navi Mumbai
5	Delhi
6	Thane
7	Chennai

	city character varying (50) 
1	Mumbai
2	Pune
3	Thane
4	Mumbai
5	Navi Mumbai
6	Pune
7	Mumbai
8	Pune

	customerid integer 
1	1
2	2

	customerid integer 
1	9
2	10

Q6

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	empid [PK] integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)
1	101	Arjun	1	38000.00	2021-06-15	[null]	arjun@novatech.com	Mumbai
2	102	Aman	1	42000.00	2022-01-10	101	aman@novatech.com	Pune
3	105	Ayaan	1	52000.00	2021-11-05	101	ayaan@novatech.com	Navi Mumbai
4	106	Anand	1	34000.00	2023-02-01	101	anand@novatech.com	Pune
5	114	Ishita	5	50000.00	2022-06-18	[null]	ishita@novatech.com	Thane
6	115	Kunal	6	45000.00	2023-01-09	[null]	kunal@novatech.com	Pune

Q7

	empid integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)	deptid integer	deptname character varying (50)	location character varying (50)	budget numeric (12,2)
1	101	Arjun	1	38000.00	2021-06-15	[null]	arjun@novatech.com	Mumbai	1	HR	Mumbai	900000.00
2	102	Aman	1	42000.00	2022-01-10	101	aman@novatech.com	Pune	1	HR	Mumbai	900000.00
3	103	Rohan	1	36000.00	2019-08-20	101	rohan@novatech.com	Thane	1	HR	Mumbai	900000.00
4	104	Kiran	1	47000.00	2020-03-12	101	kiran@novatech.com	Mumbai	1	HR	Mumbai	900000.00
5	105	Ayaan	1	52000.00	2021-11-05	101	ayaan@novatech.com	Navi Mumbai	1	HR	Mumbai	900000.00
6	106	Anand	1	34000.00	2023-02-01	101	anand@novatech.com	Pune	1	HR	Mumbai	900000.00
7	110	Vikas	3	55000.00	2018-05-14	[null]	vikas@novatech.com	Mumbai	3	Sales	Mumbai	1200000.00
8	111	Sneha	3	58000.00	2020-10-02	110	sneha@novatech.com	Pune	3	Sales	Mumbai	1200000.00
9	112	Rahul	3	62000.00	2021-01-15	110	rahul@novatech.com	Mumbai	3	Sales	Mumbai	1200000.00
10	113	Meera	4	60000.00	2019-12-12	[null]	meera@novatech.com	Navi Mumbai	4	Finance	Pune	1000000.00
11	107	Amit	2	71500.00	2019-07-11	[null]	amit@novatech.com	Mumbai	2	IT	Pune	1500000.00
12	108	Priya	2	79200.00	2021-04-19	107	priya@novatech.com	Pune	2	IT	Pune	1500000.00
13	109	Neha	2	74800.00	2022-09-23	107	neha@novatech.com	Thane	2	IT	Pune	1500000.00

Q8

	empid integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)	deptid integer	deptname character varying (50)	location character varying (50)	budget numeric (12,2)
1	101	Arjun	1	38000.00	2021-06-15	[null]	arjun@novatech.com	Mumbai	1	HR	Mumbai	900000.00
2	102	Aman	1	42000.00	2022-01-10	101	aman@novatech.com	Pune	1	HR	Mumbai	900000.00
3	103	Rohan	1	36000.00	2019-08-20	101	rohan@novatech.com	Thane	1	HR	Mumbai	900000.00
4	104	Kiran	1	47000.00	2020-03-12	101	kiran@novatech.com	Mumbai	1	HR	Mumbai	900000.00
5	105	Ayaan	1	52000.00	2021-11-05	101	ayaan@novatech.com	Navi Mumbai	1	HR	Mumbai	900000.00
6	106	Anand	1	34000.00	2023-02-01	101	anand@novatech.com	Pune	1	HR	Mumbai	900000.00
7	110	Vikas	3	55000.00	2018-05-14	[null]	vikas@novatech.com	Mumbai	3	Sales	Mumbai	1200000.00
8	111	Sneha	3	58000.00	2020-10-02	110	sneha@novatech.com	Pune	3	Sales	Mumbai	1200000.00
9	112	Rahul	3	62000.00	2021-01-15	110	rahul@novatech.com	Mumbai	3	Sales	Mumbai	1200000.00
10	113	Meera	4	60000.00	2019-12-12	[null]	meera@novatech.com	Navi Mumbai	4	Finance	Pune	1000000.00
11	114	Ishita	5	50000.00	2022-06-18	[null]	ishita@novatech.com	Thane	5	Marketing	Mumbai	800000.00
12	115	Kunal	6	45000.00	2023-01-09	[null]	kunal@novatech.com	Pune	6	Operations	Thane	700000.00

Q9

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	empid [PK] integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)
1	101	Arjun	1	38000.00	2021-06-15	[null]	arjun@novatech.com	Mumbai
2	102	Aman	1	42000.00	2022-01-10	101	aman@novatech.com	Pune
3	103	Rohan	1	36000.00	2019-08-20	101	rohan@novatech.com	Thane
4	104	Kiran	1	47000.00	2020-03-12	101	kiran@novatech.com	Mumbai
5	105	Ayaan	1	52000.00	2021-11-05	101	ayaan@novatech.com	Navi Mumbai
6	106	Anand	1	34000.00	2023-02-01	101	anand@novatech.com	Pune
7	110	Vikas	3	55000.00	2018-05-14	[null]	vikas@novatech.com	Mumbai
8	111	Sneha	3	58000.00	2020-10-02	110	sneha@novatech.com	Pune
9	112	Rahul	3	62000.00	2021-01-15	110	rahul@novatech.com	Mumbai
10	113	Meera	4	60000.00	2019-12-12	[null]	meera@novatech.com	Navi Mumbai
11	114	Ishita	5	50000.00	2022-06-18	[null]	ishita@novatech.com	Thane
12	107	Amit	2	71500.00	2019-07-11	[null]	amit@novatech.com	Mumbai
13	108	Priya	2	79200.00	2021-04-19	107	priya@novatech.com	Pune
14	109	Neha	2	74800.00	2022-09-23	107	neha@novatech.com	Thane

Q10

	productid [PK] integer	productname character varying (100)	category character varying (50)	price numeric (10,2)	stock integer
1	2	Mouse	Accessories	800.00	150
2	3	Keyboard	Accessories	1500.00	120
3	4	Monitor	Electronics	15000.00	70
4	5	Printer	Office	12000.00	60
5	6	Router	Networking	4500.00	80
6	7	Switch	Networking	6500.00	75
7	8	SSD	Storage	5500.00	100
8	9	Webcam	Accessories	3000.00	90
9	12	Projector	Office	35000.00	40
10	1	Laptop	Electronics	65000.00	40

Q20

	empid [PK] integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)
1	107	Amit	2	78650.00	2019-07-11	[null]	amit@novatech.com	Mumbai
2	108	Priya	2	87120.00	2021-04-19	107	priya@novatech.com	Pune
3	109	Neha	2	82280.00	2022-09-23	107	neha@novatech.com	Thane

Q21

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	productid [PK] integer	productname character varying (100)	category character varying (50)	price numeric (10,2)	stock integer
1	1	Laptop	Electronics	65000.00	30

Q22

	deptid [PK] integer	deptname character varying (50)	location character varying (50)	budget numeric (12,2)
1	4	Finance	Mumbai	1000000.00

Q23

	customerid [PK] integer	customername character varying (100)	city character varying (50)	country character varying (50)	phone character varying (15)	status character varying (20)
1	5	Alpha Corp	Pune	India	9999999999	ACTIVE

Q24

	empid [PK] integer	empname character varying (50)	deptid integer	salary numeric (10,2)	joiningdate date	managerid integer	email character varying (100)	city character varying (50)
1	101	Arjun	1	38000.00	2021-06-15	[null]	arjun@novatech.com	Mumbai
2	102	Aman	1	42000.00	2022-01-10	101	aman@novatech.com	Pune
3	103	Rohan	1	36000.00	2019-08-20	101	rohan@novatech.com	Thane
4	104	Kiran	1	47000.00	2020-03-12	101	kiran@novatech.com	Mumbai
5	105	Ayaan	1	52000.00	2021-11-05	101	ayaan@novatech.com	Navi Mumbai
6	106	Anand	1	34000.00	2023-02-01	101	anand@novatech.com	Pune
7	110	Vikas	3	55000.00	2018-05-14	[null]	vikas@novatech.com	Mumbai
8	111	Sneha	3	58000.00	2020-10-02	110	sneha@novatech.com	Pune
9	112	Rahul	3	62000.00	2021-01-15	110	rahul@novatech.com	Mumbai
10	113	Meera	4	60000.00	2019-12-12	[null]	meera@novatech.com	Navi Mumbai
11	114	Ishita	5	50000.00	2022-06-18	[null]	ishita@novatech.com	Thane
12	115	Kunal	6	45000.00	2023-01-09	[null]	kunal@novatech.com	Pune
13	107	Amit	2	78650.00	2019-07-11	[null]	amit@novatech.com	Mumbai
14	108	Priya	2	87120.00	2021-04-19	107	priya@novatech.com	Pune
15	109	Neha	2	82280.00	2022-09-23	107	neha@novatech.com	Thane

Q25

	orderid [PK] integer	customerid integer	productid integer	empid integer	quantity integer	orderdate date
--	--------------------------------	------------------------------	-----------------------------	-------------------------	----------------------------	--------------------------

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	productid [PK] integer	productname character varying (100)	category character varying (50)	price numeric (10,2)	stock integer
1	2	Mouse	Accessories	800.00	150
2	3	Keyboard	Accessories	1500.00	120
3	4	Monitor	Electronics	15000.00	70
4	5	Printer	Office	12000.00	60
5	6	Router	Networking	4500.00	80
6	7	Switch	Networking	6500.00	75
7	8	SSD	Storage	5500.00	100
8	9	Webcam	Accessories	3000.00	90
9	12	Projector	Office	35000.00	40
10	1	Laptop	Electronics	65000.00	30

Q26

	deptid [PK] integer	deptname character varying (50)	location character varying (50)	budget numeric (12,2)
1	1	HR	Mumbai	900000.00
2	2	IT	Pune	1500000.00
3	3	Sales	Mumbai	1200000.00
4	5	Marketing	Mumbai	800000.00
5	6	Operations	Thane	700000.00
6	4	Finance	Mumbai	1000000.00

Conclusion:

In this experiment, I successfully implemented DML operations and various SQL queries using PostgreSQL and pgAdmin 4. The results of SELECT, INSERT, UPDATE, DELETE, aggregate functions, set operations, and nested queries were verified successfully.

Signature of faculty in-charge

Department of Electronics and Telecommunication Engineering
DBMS –Sem- V- July- Nov 2026